

Task 2

Full project repo: [here](#)

A.

Describe the audio quality. Is the sound clear and continuous?

Yes, the audio sounds clear and continuous when played using the `sound()` function. There are no noticeable distortions, gaps, or interruptions. The playback faithfully represents the original recording, indicating that the file is sampled at a sufficient rate for accurate reproduction.

B.

Explain why 44.1 kHz is a standard sampling rate for audio. What is its relationship to the Nyquist Theorem?

The 44.1 kHz sampling rate is applied across the board as the norm for CD-quality sound because it meets the Nyquist Theorem while balancing storage economy against fidelity of sound. The Nyquist Theorem states that, to sample and subsequently reconstruct a signal with complete accuracy without aliasing, the sampling frequency must be at least twice the highest frequency component of the signal. Since the normal human hearing range is only up to around 20 kHz, the theoretical minimum sampling rate required is 40 kHz.

Practically, however, real-world anti-aliasing filters cannot simply cut off at exactly 20 kHz. They require some transition band in which to attenuate frequencies above the audible range. The 44.1 kHz sampling rate provides them with the buffer they require: it has enough to pass audio frequencies of up to about 20 kHz, and still has enough leeway (about 4.1 kHz) for the filter to roll off higher, unwanted frequencies before they can get a chance to produce aliasing.

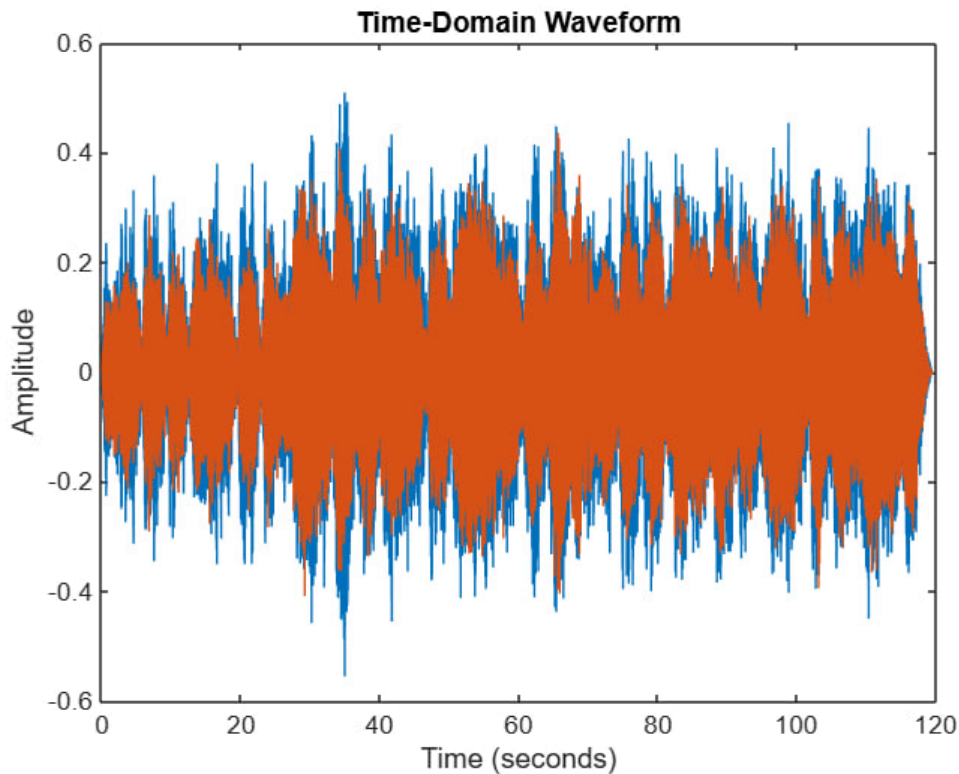
And the 44.1 kHz rate originated from the early days of digital audio and video sync needs, in which 44,100 samples per second had an excellent fit in video-based storage systems back in the 1980s. Despite new technology being available nowadays, it remains the standard due to common compatibility and adequate performance for the human ear.

In brief, 44.1 kHz satisfies the Nyquist criterion for human hearing, allows for realizable anti-aliasing filter design, and offers historically based compatibility — thereby emerging as the enduring standard for high-quality audio.

C.

How does the time-domain waveform reflect properties of the audio signal (e.g., loudness, pitch)?

This is the plot output for the audio file:



1. Loudness (Amplitude Envelope)

The loudness of the signal is mostly due to the amplitude of the signal. The larger the amplitude, the louder the sound.

The height of the waveform on the vertical axis on the time-domain plot gives its amplitude at a given moment. Wide vertical spread areas (e.g., 40–80 seconds) represent more energy and hence louder parts. Tighter areas indicate quieter sections.

Time-varying amplitude reflects dynamic changes in loudness, typical of speech, music, or field recordings of ambient sounds.

2. Pitch (Frequency Content)

Pitch is concerned with the frequency of oscillations in the signal. Increasing frequencies map onto increasing pitches perceived.

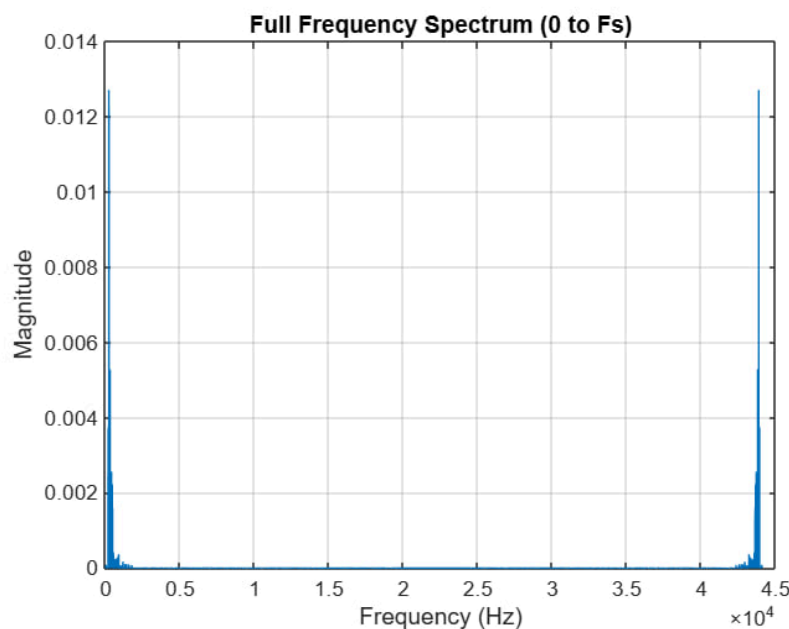
Graphically, areas of closely spaced oscillations are areas of high frequency components, and areas of slowly moving, wider oscillations are lower frequencies.

The density of the waveform is a rough guide to pitch change over time, but accurate frequency analysis needs to use spectral methods.

D.

What insights about the music's frequency content does the Fourier analysis provide? Compare this with the time-domain view.

This is the plot output:



As we saw, the time-domain representation of the audio signal demonstrates how its amplitude changes over time. This representation provides valuable information about:

- **Loudness & Dynamics:** The peaks of the waveform indicate the louder sections (such as drum attacks, vocal peaks, or climaxes), while the troughs indicate quieter sections or silence. This reveals the overall energy profile of the track.

- **Rhythmic and Temporal Structure:** Rhythm, timing, and phrasing in the music are defined by the waveform. Within the waveform patterns, repeating motifs, emphasis on the beat, and musical transitions are marked.
- **Transient and Envelope Features:** Sudden changes or spikes represent transients (e.g., percussion), whereas smooth transitions represent gradual musical progressions like sustained tones or ambient sections.

Meanwhile, the Fourier-transformed magnitude spectrum reveals the frequency content of the audio signal, and which frequencies are most prominent:

- **Low-Frequency Dominance:** Energy of the signal is mostly below ~2500 Hz, indicating strong presence of bass—such as bass instruments, kick drums, or male vocals. It typically forms the bottom layer of the majority of musical genres.
- **Spectral Sparsity at Midrange:** The mid-frequency band appears less dominant, possibly on account of stylistic choice (e.g., zero harmonic piling) or production equalization used.
- **High-Frequency Peaks:** A small spike near the Nyquist frequency (~22.05 kHz on 44.1 kHz sampling) can indicate aliasing artifacts, reflection of real-valued signal symmetry or digital processing issues such as DC offset or poor filtering.
- **Filtering Effects:** Certain of the fundamental frequencies sound subdued, suggesting that low-pass, band-stop, or notch filtering was deliberately applied at the point of mixing or mastering.

When combined, the time-domain and frequency-domain plots create a holistic description of the audio signal:

The time-domain plot helps to identify where things happen—such as rhythm, when loudness varies, and musical phrasing.

Plot in the frequency domain helps identify what is inside the signal—such as pitch content, harmonic richness, or the presence of specific instruments.

Comparable, cyclical, high-energy structures in both domains point to loop-based or beat-driven composition—common in electronic, hip-hop, or pop music.

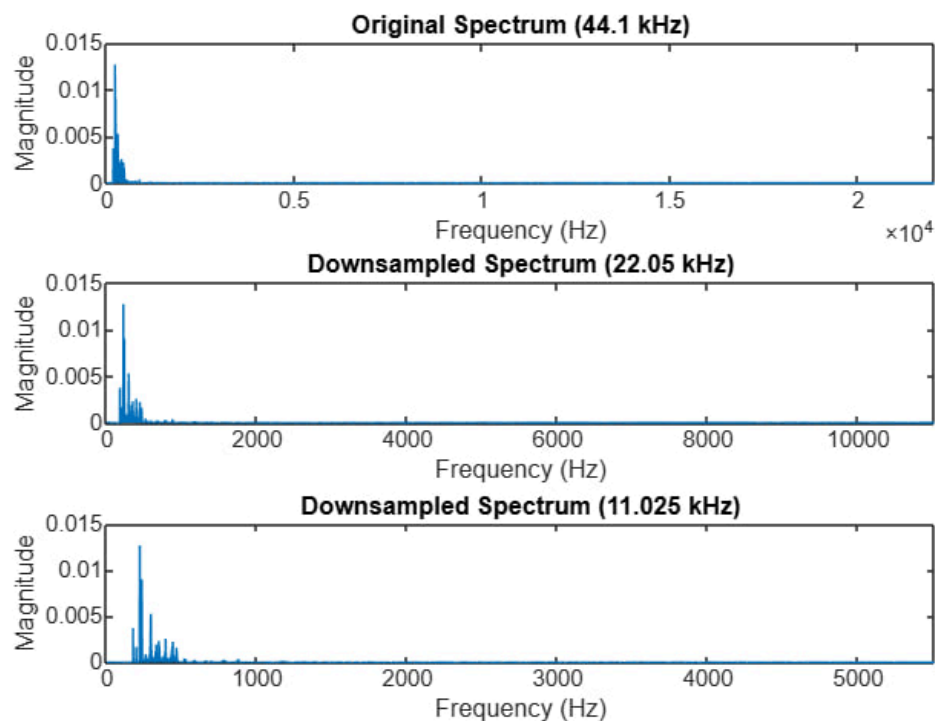
The distribution and ratio of energy over the spectrum reflect stylistic choice and technical aspects of the audio production process.

This concomitant consideration of time-domain and frequency-domain representations allows for deeper understanding of the music's structure, content, and character. While the waveform outlines how the sound evolves over time, the spectral view discloses its harmonic and tonal structure.

E.

How does reducing the sampling rate affect the audio quality and Fourier analysis results?

This is the plot output for each downsampling:



According to the Nyquist Theorem, the highest frequency that can be accurately represented in a digital signal is half the sampling rate. For standard audio with 44.1 kHz sampling, frequencies up to 22.05 kHz can be represented. As the audio is downsampled to lower frequencies like 22.05 kHz and 11.025 kHz, the Nyquist frequency drops to 11.025 kHz and 5.5125 kHz, respectively.

Thus, all high-frequency content above these thresholds is rejected or aliased, and audio detail is compromised. Clarity, brightness, and sound sharpness are particularly affected by this. For example, cymbals, harmonics, or any subtle high-end textures become blurred or completely lost, leading to a noticeably duller or poorer quality playback in the time domain.

In a spectral (Fourier) analysis situation, this reducing of the sample rate leads to:

- A cutoff in the frequency spectrum at the new Nyquist frequency.
- Fewer visible peaks in the high-frequency range.
- A more compressed spectrum, since higher harmonics are no longer being recorded.

These effects directly demonstrate the trade-off between audio fidelity and data size. Decreasing the sampling rates reduces file size at the cost of audible quality and frequency resolution. While this may be acceptable for speech or low-bandwidth applications, it is unacceptable for high-quality music reproduction or for detailed audio analysis.